

Question	Marking Guidance	Mark	Comments
06.1	1. The shape of the active site becomes <u>complementary</u> (to the hydrogen peroxide/ substrate); 2. (So) tension/strain/twisting/bending of the bond(s) (in hydrogen peroxide); 3. (So) less energy needed to break the bond(s) OR Lowers activation energy;	3 (3 x AO1)	1. Reject catalase has similar/same shape as hydrogen peroxide 2. Ignore 'weakens bonds'

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06.2	1. Label glassware/beaker/flasks/boiling tubes/hydrogen peroxide because hydrogen peroxide could be mistaken for water; 2. Wear gloves to prevent hydrogen peroxide/solution making contact with skin/damaging skin; 3. (Work in) a well-ventilated room or fume cupboard to prevent inhaling hydrogen peroxide (fumes) OR to prevent respiratory damage; 4. Wear (safety) glasses/goggles to prevent hydrogen peroxide entering the eyes/damaging the eyes;	2 max (2 x AO2)	Each mark needs precaution and reason 2. Accept lab coat or mask/visor but only if idea of covering the skin is clear 3. Allow mask

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06.3	To expose all of the surface (to the hydrogen peroxide/substrate) OR To maximise the surface exposed (to the hydrogen peroxide);	1 (1 x AO2)	Ignore increasing the surface area

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06.4	Correct answer of $1.3/1.25 = 3$ marks;;; Incorrect answer but shows 3140 and 2512 = 2 marks OR Incorrect answer but shows 3142 and 2513 = 2 marks OR Incorrect answer of $1.5/1.47 = 2$ marks;; Incorrect answer but shows $628/628.3 = 1$ mark;	3 (3 x AO2)	Surface areas 3140 and 2512 if 3.14 used for π Surface areas 3142 and 2513 if calculator used for π 0.8 if surface area of cylinder divided by surface area of discs 1.5/1.47 if diameter used instead of radius 628 for surface area of one disc
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06.5	(Total surface area greater so) more enzyme-substrate complexes;	1 (1 x AO2)	Accept E-S complexes